

HDS06 - Designing and Querying a Clinical Trial Participant Registry

Westbrook University Hospitals NHS Trust, Clinical Trial Participant Registry

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Introduction

This report designs a fictional relational database for the Westbrook University Hospitals NHS Trust Clinical Trial Participant Registry. It implements it in a Microsoft SQL Server (T-SQL), populates it and answers seven fictional clinical questions. The accompanying runnable script is `script/HDS_DB_LAINSBURY_2026.sql`.

Note

Reproducing the results. Run `script/HDS_DB_LAINSBURY_2026.sql`. It the requested single self-contained script that creates the database, tables, data, trigger and queries in order. It ran on my instance without errors from start to finish. Each query returns one result grid that are screenshotted below. I used the VS Code mssql extension connected to a local SQL Server.

Chapter 1

Part 1: Database Design

1.1 Entity Relationship Diagram

The DB schema has eight tables. **TrialSite** is necessary addition that resolves the many-to-many relationship between trials and sites. **Enrollment** records a participant's participation in a trial at a site. **ConsentSigning** is the append-only log of consent events to ensure a complete audit trail. The ERD is shown in crow's-foot notation in Figure 1.1.

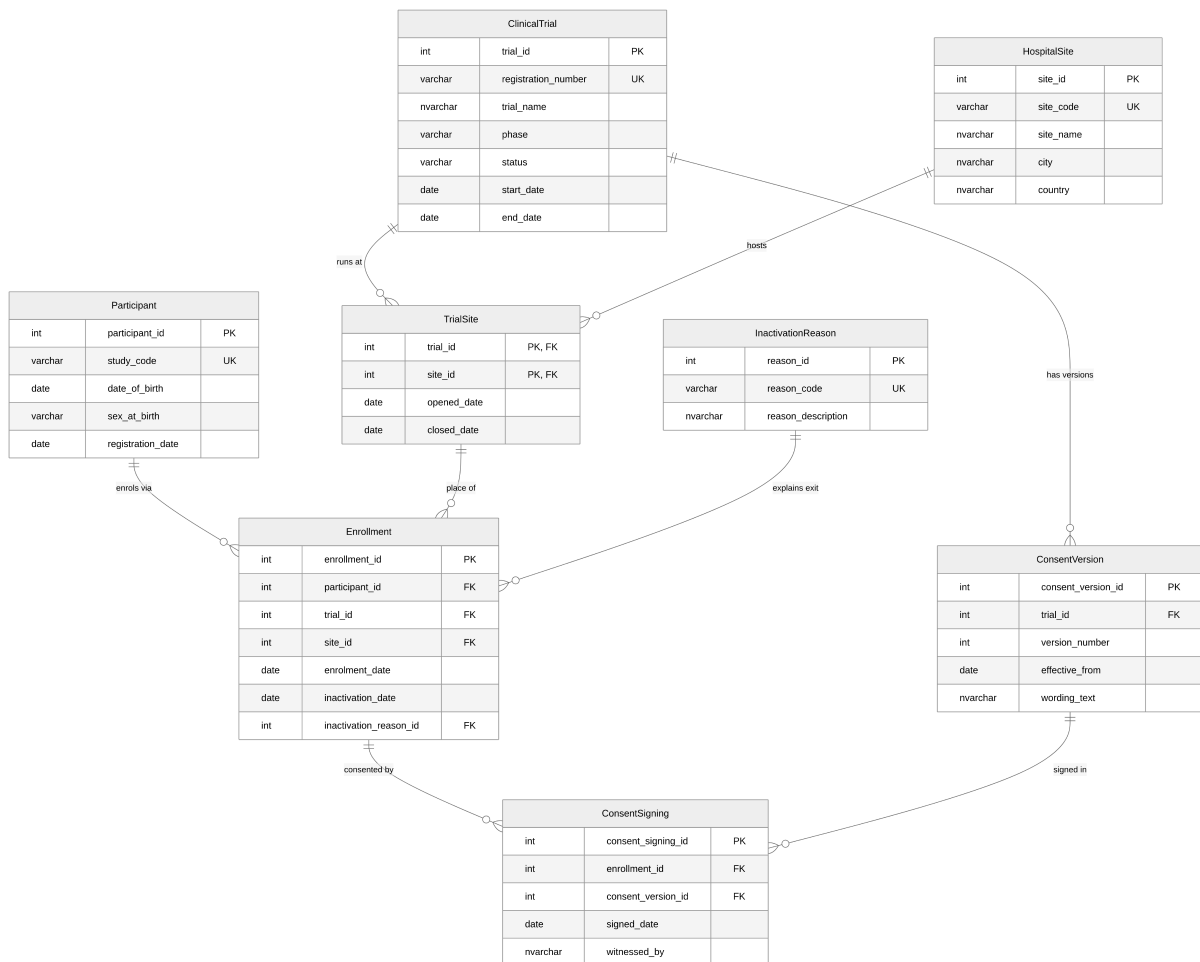


Figure 1.1: Entity Relationship Diagram (crow's-foot notation).

1.2 Table Descriptions

Every table uses a surrogate INT **IDENTITY** primary key for internal joins, plus one or more **business UNIQUE keys** for the values clinical staff actually recognise.

Table 1.1: Table descriptions and key design choices

Table	Purpose and key design choices
ClinicalTrial	One row per trial. PK <code>trial_id</code> (IDENTITY). UNIQUE registration_number — the ISRCTN number is the externally recognised, never-reused identifier, so it is the natural key. CHECK on <code>phase</code> (I/II/III/IV) and <code>status</code> enforce the controlled vocabularies; CHECK (<code>end_date >= start_date</code>) prevents impossible ranges.
HospitalSite	One row per site. PK <code>site_id</code> . UNIQUE site_code — short codes (e.g. WBK-GEN) are how sites are referred to operationally, so they form the natural key.
Participant	One pseudonymised participant per row; no names held . PK <code>participant_id</code> . UNIQUE study_code — the pseudonym is the only safe external identifier. CHECK on <code>sex_at_birth</code> (biological sex at birth, not gender identity — only two values are recorded); CHECK (<code>date_of_birth < registration_date</code>).
InactivationReason	Controlled vocabulary of reasons a participant leaves a trial, so exits report consistently across trials. PK <code>reason_id</code> . UNIQUE reason_code — the stable report token.
TrialSite	Link table resolving the trial–site many-to-many; also records when each site opened/closed for recruitment on that trial. Composite PK (<code>trial_id</code> , <code>site_id</code>) — the pairing itself is the row’s identity, so no surrogate IDENTITY column is used here (unlike every other table); FKs to ClinicalTrial and HospitalSite ; CHECK (<code>closed_date >= opened_date</code>).
ConsentVersion	Every approved version of a trial’s consent form, including wording and effective date; old versions never overwritten. PK <code>consent_version_id</code> . UNIQUE (<code>trial_id</code> , <code>version_number</code>); FK to ClinicalTrial ; <code>wording_text</code> is NVARCHAR(MAX) .
Enrollment	A participant enrolled in a trial at a specific site. Carries (<code>trial_id</code> , <code>site_id</code>) as a composite FK straight to TrialSite , so trial and site are captured together and a participant can only enrol where the trial actually runs. Exit is recorded without deleting the row. PK <code>enrollment_id</code> ; FKs to Participant , TrialSite (composite), InactivationReason ; UNIQUE (<code>participant_id</code> , <code>trial_id</code> , <code>site_id</code>); paired-null CHECK ; filtered unique index for “one active trial at a time”.
ConsentSigning	Append-only log of every consent signing event: when, which version, who witnessed. PK <code>consent_signing_id</code> ; FKs to Enrollment and ConsentVersion ; UNIQUE (<code>enrollment_id</code> , <code>consent_version_id</code>); protected by an append-only trigger.

Check constraints.

- Phase/status/sex use `CHECK ... IN (...)` to guarantee clean categorical data for cross-trial reporting.
- Date checks (`end_date >= start_date`, `closed_date >= opened_date`, `inactivation_date >= enrolment_date`) prevent logically impossible records.
- The paired-null check on `Enrollment` guarantees a leaving date never appears without a reason and vice-versa.

1.3 NULL and NOT NULL Decisions

Table 1.2: Nullability decisions and their justification

Column	Decision	Rationale
<code>ClinicalTrial.end_date</code>	NULL	NULL is meaningful: the trial is still running. Current trials have no end date <i>yet</i> .
<code>ClinicalTrial.start_date</code> , <code>status</code> , <code>phase</code>	NOT NULL	A trial cannot exist without these.
<code>TrialSite.closed_date</code>	NULL	The site can still be open for recruitment on that trial.
<code>Enrollment.inactivation_date</code> , <code>inactivation_reason_id</code>	NULL (paired)	The classic “meaningful absence”. The paired-null <code>CHECK</code> keeps them in step.
<code>Participant.date_of_birth</code> , <code>registration_date</code> , <code>sex_at_birth</code>	NOT NULL	Required for eligibility, age and demographic reporting.
<code>ConsentSigning.signed_date</code> , <code>witnessed_by</code>	NOT NULL	A consent event without a date or witness is not a valid consent event.
<code>ConsentVersion.wording_text</code> , <code>effective_from</code>	NOT NULL	A consent version is meaningless without its wording and the date it took effect.

1.4 Relationships Between Tables

Table 1.3: Relationships between tables

Tables	Type / FK	What it represents
<code>ClinicalTrial</code> – <code>HospitalSite</code>	M:N via <code>TrialSite</code> (FK <code>trial_id</code> , <code>site_id</code>)	A trial runs at many sites and a site hosts many trials. The link table also carries the open/close recruitment dates.
<code>ClinicalTrial</code> -> <code>ConsentVersion</code>	1:M (FK <code>trial_id</code>)	One trial accumulates many consent-form versions over its life as the protocol changes.

Tables	Type / FK	What it represents
Participant -> Enrollment	1:M (FK <code>participant_id</code>)	A participant may have several enrolments over time, only one active at once as defined per brief (filtered index).
TrialSite -> Enrollment	1:M (composite FK <code>trial_id, site_id</code>)	Each enrolment happens at exactly one trial-site pairing that the TrialSite link table confirms actually exists.
InactivationReason -> Enrollment	1:M optional (FK <code>inactivation_reason_id</code>)	A standardised reason classifies many exits; active enrolments reference none.
Enrollment -> ConsentSigning	1:M (FK <code>enrollment_id</code>)	One enrolment can have several signing events (original + re-consent after amendment).
ConsentVersion -> ConsentSigning	1:M (FK <code>consent_version_id</code>)	Each signing records which version was signed; one version is signed by many participants.

Many-to-many resolution. The only many-to-many relationship (ClinicalTrial – HospitalSite) is resolved with the **TrialSite** junction table. I decided to use a composite primary key (`trial_id, site_id`). Therefore no surrogate is needed since the pairing itself is the row's identity.

Chapter 2

Part 2: SQL

2.1 INSERT statements

Every INSERT that needs to reference another table (e.g. Enrollment needs a trial_id) looks up the ID up by a human-recognisable business key instead of hardcoding the raw number. There are no hard-coded IDENTITY integers to avoid mistakes.

Note

I have only shown the first insert per table in this report to illustrate the pattern. All INSERTs are in script/HDS_DB_LAINSBURY_2026.sql.

ClinicalTrial

```
INSERT INTO dbo.ClinicalTrial (registration_number, trial_name, phase, status,
↪ start_date, end_date)
VALUES ('ISRCTN10234567', N'CARDIOPROTECT: Cardiac Rehabilitation in Post-MI
↪ Patients',
        'III', 'active', '2023-03-01', NULL);
```

HospitalSite

```
INSERT INTO dbo.HospitalSite (site_code, site_name, city, country)
VALUES ('WBK-GEN', N'Westbrook General Hospital', N'London', N'England');
```

Participant

```
INSERT INTO dbo.Participant (study_code, date_of_birth, sex_at_birth,
↪ registration_date)
VALUES ('WBK-001', '1958-04-12', 'Male', '2021-11-03');
```

InactivationReason

```
INSERT INTO dbo.InactivationReason (reason_code, reason_description)
VALUES ('consent_withdrawn', N'Participant withdrew their consent to take part
↳ in the trial');
```

TrialSite (FK look-ups via subquery)

```
INSERT INTO dbo.TrialSite (trial_id, site_id, opened_date, closed_date)
VALUES (
  (SELECT trial_id FROM dbo.ClinicalTrial WHERE
↳ registration_number='ISRCTN10234567'),
  (SELECT site_id FROM dbo.HospitalSite WHERE site_code='WBK-GEN'),
  '2023-03-01', NULL);
```

ConsentVersion

```
INSERT INTO dbo.ConsentVersion (trial_id, version_number, effective_from,
↳ wording_text)
VALUES (
  (SELECT trial_id FROM dbo.ClinicalTrial WHERE
↳ registration_number='ISRCTN10234567'),
  1, '2023-03-01',
  N'I agree to take part in the CARDIOPROTECT study. ... I may withdraw at any
↳ time without affecting my medical care.');
```

Enrollment (resolves trial_id and site_id directly from business keys)

```
INSERT INTO dbo.Enrollment (participant_id, trial_id, site_id, enrolment_date,
↳ inactivation_date, inactivation_reason_id)
VALUES (
  (SELECT participant_id FROM dbo.Participant WHERE study_code='WBK-001'),
  (SELECT trial_id FROM dbo.ClinicalTrial WHERE
↳ registration_number='ISRCTN10234567'),
  (SELECT site_id FROM dbo.HospitalSite WHERE site_code='WBK-GEN'),
  '2023-03-15', NULL, NULL);
```

ConsentSigning (resolves enrolment + version by business keys)

```
INSERT INTO dbo.ConsentSigning (enrollment_id, consent_version_id, signed_date,
    witnessed_by)
VALUES (
    (SELECT e.enrollment_id FROM dbo.Enrollment e
        JOIN dbo.Participant p ON p.participant_id = e.participant_id
        JOIN dbo.ClinicalTrial ct ON ct.trial_id = e.trial_id
        WHERE p.study_code='WBK-001' AND ct.registration_number='ISRCTN10234567'),
    (SELECT cv.consent_version_id FROM dbo.ConsentVersion cv
        JOIN dbo.ClinicalTrial ct ON ct.trial_id = cv.trial_id
        WHERE ct.registration_number='ISRCTN10234567' AND cv.version_number=1),
    '2023-03-15', N'Nurse A. Okafor');
```

2.2 SQL Queries

2.2.1 Query 1: Current consent wording for CARDIOPROTECT

Clinical question. Before a consent discussion, a research nurse needs the current (most recently effective) consent wording for CARDIOPROTECT: version number, effective date, and full wording.

```
SELECT TOP (1) cv.version_number, cv.effective_from, cv.wording_text
FROM    dbo.ConsentVersion cv
JOIN    dbo.ClinicalTrial ct ON ct.trial_id = cv.trial_id
WHERE   ct.trial_name LIKE 'CARDIOPROTECT%'
ORDER  BY cv.effective_from DESC;
```

Screenshot.

```

2  USE HDS_ClinicalTrial;
3  GO
4
5  /*
6   QUERY 1
7   Current (most recently effective) consent wording for the CARDIOPROTECT trial. */
8  SELECT TOP (1)
9      cv.version_number,
10     cv.effective_from,
11     cv.wording_text
12 FROM     dbo.ConsentVersion cv
13 JOIN     dbo.ClinicalTrial ct ON ct.trial_id = cv.trial_id
14 WHERE    ct.trial_name LIKE 'CARDIOPROTECT%'
15 ORDER BY cv.effective_from DESC;
16 GO

```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

QUERY RESULTS

Results (1)

Messages

	version_number	effective_from	wording_text
1 2		2024-01-15	I agree to take part in the CARDIOPROTECT stu...

Figure 2.1: Query 1 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. The query filters to CARDIOPROTECT, orders versions by `effective_from` descending and keeps the top row, so it always returns the latest version even after future amendments. *Data quality:* it relies on `effective_from` being accurate, a new version entered with a wrong or missing effective date could show the nurse out-of-date wording.

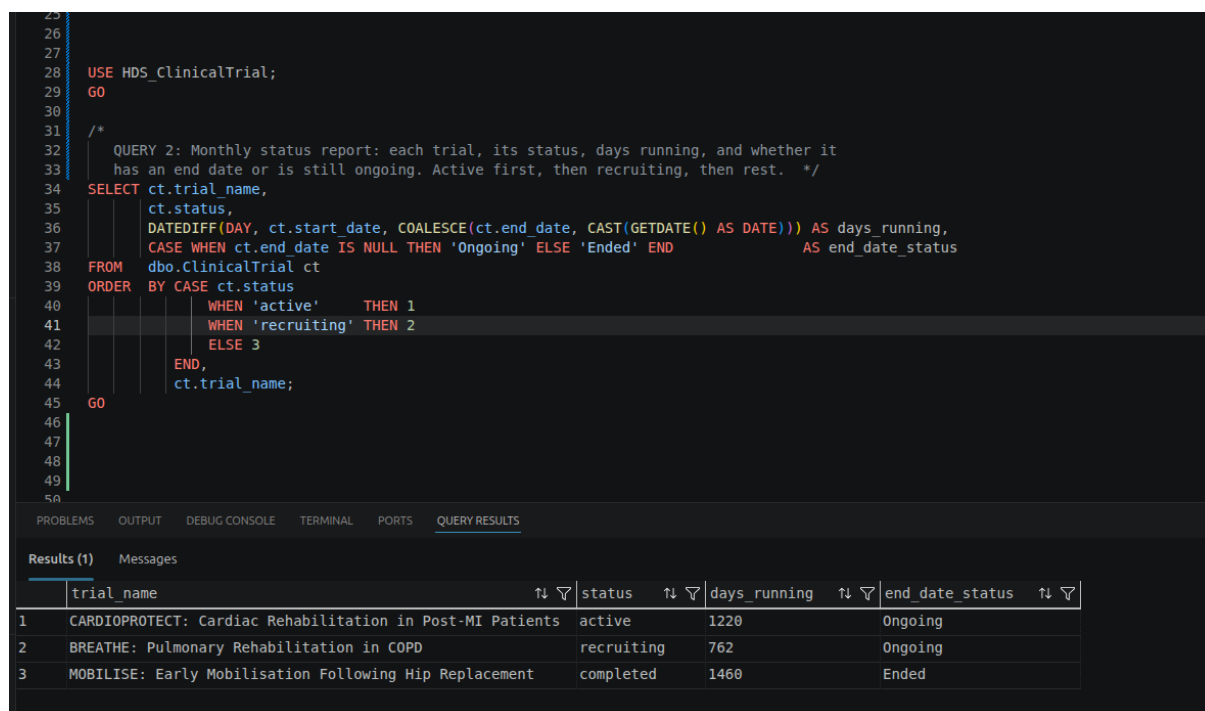
2.2.2 Query 2: Monthly trial status report

Clinical question. A summary of each trial, its status, how long it has run in days, and whether it has an end date or is ongoing; active trials first, then recruiting, then the rest.

```
SELECT ct.trial_name, ct.status,
       DATEDIFF(DAY, ct.start_date, COALESCE(ct.end_date, CAST(GETDATE() AS
    DATE))) AS days_running,
       CASE WHEN ct.end_date IS NULL THEN 'Ongoing' ELSE 'Ended' END AS
    end_date_status
FROM    dbo.ClinicalTrial ct
ORDER  BY CASE ct.status WHEN 'active' THEN 1 WHEN 'recruiting' THEN 2 ELSE 3
    END,
```

```
ct.trial_name;
```

Screenshot.



```

25
26
27
28 USE HDS_ClinicalTrial;
29 GO
30
31 /*
32  QUERY 2: Monthly status report: each trial, its status, days running, and whether it
33  has an end date or is still ongoing. Active first, then recruiting, then rest. */
34 SELECT ct.trial_name,
35        ct.status,
36        DATEDIFF(DAY, ct.start_date, COALESCE(ct.end_date, CAST(GETDATE() AS DATE))) AS days_running,
37        CASE WHEN ct.end_date IS NULL THEN 'Ongoing' ELSE 'Ended' END           AS end_date_status
38 FROM    dbo.ClinicalTrial ct
39 ORDER BY CASE ct.status
40           WHEN 'active'      THEN 1
41           WHEN 'recruiting' THEN 2
42           ELSE 3
43           END,
44        ct.trial_name;
45 GO
46
47
48
49
50

```

	trial_name	status	days_running	end_date_status
1	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	active	1220	Ongoing
2	BREATHE: Pulmonary Rehabilitation in COPD	recruiting	762	Ongoing
3	MOBILISE: Early Mobilisation Following Hip Replacement	completed	1460	Ended

Figure 2.2: Query 2 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. COALESCE substitutes today’s date for trials with no end date so “days running” is meaningful for ongoing trials, and a CASE expression in ORDER BY gives the custom status priority. *Data quality:* “days running” for ongoing trials changes every day the report is run, so it is only a dated snapshot.

2.2.3 Query 3: Trial history for participant WBK-003

Clinical question. Full history for WBK-003: every trial, enrolment date, inactivation date, and reason code; ordered by enrolment date. Active enrolments have no reason.

```

SELECT ct.trial_name, e.enrolment_date, e.inactivation_date, ir.reason_code
FROM    dbo.Enrollment      e
JOIN    dbo.Participant      p ON p.participant_id = e.participant_id
JOIN    dbo.ClinicalTrial    ct ON ct.trial_id      = e.trial_id
LEFT JOIN dbo.InactivationReason ir ON ir.reason_id =
    e.inactivation_reason_id
WHERE   p.study_code = 'WBK-003'
ORDER BY e.enrolment_date;

```

Screenshot.

```

51
52
53
54 USE HDS_ClinicalTrial;
55 GO
56
57 /*
58  QUERY 3: Full trial history for participant WBK-003. A LEFT JOIN to InactivationReason
59  keeps active enrolments (which have no reason) in the result set. */
60 SELECT ct.trial_name,
61        e.enrolment_date,
62        e.inactivation_date,
63        ir.reason_code
64 FROM    dbo.Enrollment      e
65 JOIN    dbo.Participant     p ON p.participant_id = e.participant_id
66 JOIN    dbo.TrialSite      ts ON ts.trial_site_id = e.trial_site_id
67 JOIN    dbo.ClinicalTrial  ct ON ct.trial_id      = ts.trial_id
68 LEFT JOIN dbo.InactivationReason ir ON ir.reason_id = e.inactivation_reason_id
69 WHERE   p.study_code = 'WBK-003'
70 ORDER BY e.enrolment_date;
71 GO
72
73
74
75

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	trial_name	enrolment_date	inactivation_date	reason_code
1	BREATHE: Pulmonary Rehabilitation in COPD	2024-06-10	2024-09-15	consent_withdrawn
2	CARDIOPROTECT: Cardiac Rehabilitation in Post...	2024-10-01	NULL	NULL

Figure 2.3: Query 3 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. A LEFT JOIN to InactivationReason is essential: an active enrolment has a NULL reason and an INNER join would silently drop it. Enrollment stores trial_id directly (as part of its composite FK to TrialSite), so ClinicalTrial is joined straight off it with no TrialSite hop needed.

Data quality: a participant who left but whose inactivation_reason_id was never populated would show a leaving date with a blank reason, the paired-null check constraint I entered in the table prevents this.

2.2.4 Query 4: Audit of currently active enrolments

Clinical question. Each currently **active** participant with their trial, site, and enrolment date; ordered by trial name then enrolment date.

```

SELECT p.study_code, ct.trial_name, hs.site_name, e.enrolment_date
FROM    dbo.Enrollment      e
JOIN    dbo.Participant     p ON p.participant_id = e.participant_id
JOIN    dbo.ClinicalTrial  ct ON ct.trial_id      = e.trial_id
JOIN    dbo.HospitalSite   hs ON hs.site_id       = e.site_id
WHERE   e.inactivation_date IS NULL
ORDER BY ct.trial_name, e.enrolment_date;

```

Screenshot.

```

77
78
79
80 USE HDS_ClinicalTrial;
81 GO
82 /*
83  QUERY 4: Audit of currently ACTIVE enrolments: participant, trial, site, enrol date.
84  Order by trial name then enrolment date. */
85 SELECT p.study_code,
86        ct.trial_name,
87        hs.site_name,
88        e.enrolment_date
89 FROM   dbo.Enrollment e
90 JOIN   dbo.Participant p ON p.participant_id = e.participant_id
91 JOIN   dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
92 JOIN   dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
93 JOIN   dbo.HospitalSite hs ON hs.site_id = ts.site_id
94 WHERE  e.inactivation_date IS NULL
95 ORDER BY ct.trial_name, e.enrolment_date;
96 GO
97
98
99
100
101
102

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	study_code	trial_name	site_name	enrolment_date
1	WBK-005	BREATHE: Pulmonary Rehabilitation in COPD	Westbrook General Hospital	2024-08-05
2	WBK-001	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	Westbrook General Hospital	2023-03-15
3	WBK-003	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	Westbrook General Hospital	2024-10-01
4	WBK-002	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	St Katherine's Hospital	2025-01-10

Figure 2.4: Query 4 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. “Active” is expressed as `inactivation_date IS NULL`. Enrollment stores `trial_id` and `site_id` directly, so `ClinicalTrial` and `HospitalSite` are joined straight off it, no `TrialSite` hop needed to reconstruct where each active participant sits.

Data quality: the definition of “active” depends on inactivation being recorded promptly. A participant who has left but whose exit was not entered would wrongly appear here.

2.2.5 Query 5: Enrolment count per site (including empty sites)

Clinical question. How many participants have ever been enrolled at each site, including sites with none. Show site name, city, total count, descending.

```

SELECT hs.site_name, hs.city, COUNT(e.enrollment_id) AS total_enrolments
FROM   dbo.HospitalSite hs
LEFT   JOIN dbo.Enrollment e ON e.site_id = hs.site_id
GROUP BY hs.site_name, hs.city
ORDER BY total_enrolments DESC;

```

Screenshot.

```

99
100
101 USE HDS_ClinicalTrial;
102 GO
103
104 /*
105  QUERY 5: Total enrolments ever, per site, INCLUDING sites with none.
106  LEFT JOIN from HospitalSite preserves sites with zero enrolments;
107  COUNT(e.enrollment_id) counts rows, not NULLs, so empty sites score 0.*/
108 SELECT hs.site_name,
109        hs.city,
110        COUNT(e.enrollment_id) AS total_enrolments
111 FROM    dbo.HospitalSite hs
112 LEFT JOIN dbo.TrialSite ts ON ts.site_id = hs.site_id
113 LEFT JOIN dbo.Enrollment e ON e.trial_site_id = ts.trial_site_id
114 GROUP BY hs.site_name, hs.city
115 ORDER BY total_enrolments DESC;
116 GO
117
118
119
120
121
122
123

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	site_name	city	total_enrolments
1	Westbrook General Hospital	London	4
2	St Katherine's Hospital	Manchester	3
3	Ridgeway Community Hospital	Bristol	0

Figure 2.5: Query 5 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. Starting from `HospitalSite` with a `LEFT JOIN` preserves sites with no enrolments, and `COUNT(e.enrollment_id)` counts rows rather than `NULLs` so empty sites correctly score 0. `Enrollment` stores `site_id` directly, so it is joined straight off `HospitalSite` with no `TrialSite` hop needed.

Data quality: A site that participates in a trial but has not yet recorded any enrolments is indistinguishable from one that genuinely has none, the zero needs interpretation in context.

2.2.6 Query 6: Trials with an amended protocol

Clinical question. Trials where the protocol was amended (more than one consent version): trial name, current status, number of versions; descending; only where more than one version exists.

```

SELECT ct.trial_name, ct.status, COUNT(cv.consent_version_id) AS version_count
FROM    dbo.ClinicalTrial ct
JOIN    dbo.ConsentVersion cv ON cv.trial_id = ct.trial_id
GROUP BY ct.trial_name, ct.status
HAVING COUNT(cv.consent_version_id) > 1
ORDER BY version_count DESC;

```

Screenshot.


```

121
122
123
124 USE HDS_ClinicalTrial;
125 GO
126 /*
127  QUERY 6: Trials with an amended protocol (more than one consent version issued). */
128 SELECT ct.trial_name,
129        ct.status,
130        COUNT(cv.consent_version_id) AS version_count
131 FROM    dbo.ClinicalTrial ct
132 JOIN    dbo.ConsentVersion cv ON cv.trial_id = ct.trial_id
133 GROUP BY ct.trial_name, ct.status
134 HAVING COUNT(cv.consent_version_id) > 1
135 ORDER BY version_count DESC;
136 GO
137
138
139
140
141
142
143
144
145

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	trial_name	status	version_count
1	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	active	2

Figure 2.6: Query 6 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. Grouping by trial and counting versions, then filtering with `HAVING COUNT(...) > 1`, isolates amended trials (`HAVING` filters the aggregate, which `WHERE` cannot).

Data quality: this counts version *rows*, so a duplicate version entered in error would overstate the number of genuine amendments.

2.2.7 Query 7: Most recently enrolled participant per site

Clinical question. The most recently enrolled participant at each site across all trials: study code, trial name, site name, enrolment date.

```

WITH RankedEnrolments AS (
    SELECT p.study_code, ct.trial_name, hs.site_name, e.enrolment_date,
           ROW_NUMBER() OVER (PARTITION BY hs.site_id
                               ORDER BY e.enrolment_date DESC, e.enrollment_id
                                     DESC) AS rn
    FROM    dbo.Enrollment e
    JOIN    dbo.Participant p ON p.participant_id = e.participant_id
    JOIN    dbo.ClinicalTrial ct ON ct.trial_id = e.trial_id
    JOIN    dbo.HospitalSite hs ON hs.site_id = e.site_id
)
SELECT study_code, trial_name, site_name, enrolment_date
FROM    RankedEnrolments
WHERE   rn = 1
ORDER BY site_name;

```

Screenshot.

```

151 USE HDS_ClinicalTrial;
152 GO
153 /* QUERY 7: Most recently enrolled participant at each site (across all trials).
154    ROW_NUMBER partitioned by site, newest enrolment first; keep rank 1. */
155 WITH RankedEnrolments AS (
156     SELECT p.study_code,
157            ct.trial_name,
158            hs.site_name,
159            e.enrolment_date,
160            ROW_NUMBER() OVER (PARTITION BY hs.site_id
161                               ORDER BY e.enrolment_date DESC, e.enrollment_id DESC) AS rn
162     FROM   dbo.Enrollment e
163     JOIN   dbo.Participant p ON p.participant_id = e.participant_id
164     JOIN   dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
165     JOIN   dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
166     JOIN   dbo.HospitalSite hs ON hs.site_id = ts.site_id
167 )
168 SELECT study_code,
169        trial_name,
170        site_name,
171        enrolment_date
172 FROM   RankedEnrolments
173 WHERE  rn = 1
174 ORDER BY site_name;
175 GO
176

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	study_code	trial_name	site_name	enrolment_date
1	WBK-002	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	St Katherine's Hospital	2025-01-10
2	WBK-003	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	Westbrook General Hospital	2024-10-01

Figure 2.7: Query 7 output grid (VS Code mssql extension, SQL Server 2025).

Discussion. `ROW_NUMBER()` partitioned by site and ordered by enrolment date descending ranks each site’s enrolments; keeping `rn = 1` returns the newest per site, with `enrollment_id` as a deterministic tie-breaker. `Enrollment` stores `trial_id` and `site_id` directly, so `ClinicalTrial` and `HospitalSite` are joined straight off it with no `TrialSite` hop needed.

Data quality: if two participants share the same enrolment date the tie-break is arbitrary, so “most recent” may need a finer timestamp than a date to be unambiguous.

Chapter 3

Part 3: Reflection

3.1 Why no participant names?

Storing only a `study_code` pseudonymises the registry. This follows the data-minimisation principle from the GDPR *Principles relating to processing of personal data*¹ (GDPR Article 5(1)(c)).

Additional, we could also refer to the Caldicott Principles² of justifying every use of confidential information and using the minimum necessary which reinforces the article 5 of the GDPR. This means a breach of the trial database does not directly expose identities and identities does not bring necessary information to the database.

Direct identifiers such as names would live in a separate, access-controlled master linkage list held by the research office, not in this analytical database. Researchers can query and report entirely on `study_code`, so trial analyses can be shared more freely because they carry no obvious identifiers. The cost is friction and a dependency on the linkage being accurate and available: a broken or out-of-date mapping could mean a participant cannot be reached for a safety issue. The design therefore trades day-to-day convenience for a much stronger confidentiality and information-governance position.

3.2 A situation where the data could mislead

Consider Query 4 and Query 5, which both depend on `inactivation_date`. If a participant has actually left a trial but their exit has not yet been entered, they will still appear as “active” and will be counted in active-enrolment reports — overstating recruitment and potentially prompting unnecessary follow-up contact. The data is not *wrong* field by field; it is *stale*, which is harder to spot. A data manager could detect this by reconciling the registry against source records: cross-checking active enrolments against clinic attendance or against the standardised `InactivationReason` log, and flagging participants with no recent activity (e.g. no consent or visit events for an unusually long period) for review. Preventively, they could add a periodic data-quality report listing long-inactive “active” enrolments, set a service standard for how quickly exits must be recorded, and use the `lost_to_follow_up` reason code consistently so genuine non-contact is captured rather than left as a silent gap. The fix is process plus monitoring,

not just a query.

3.3 Why consent records are append-only, and how to enforce it

Consent records are the legal and ethical evidence that each participant agreed to the specific protocol version in force **at the time**. They must be append-only because consent is a historical fact: Good Clinical Practice (ICH-GCP) requires trial-critical records to carry a complete, tamper-evident audit trail, and editing or deleting a signing event would destroy the evidence that regulators, sponsors and ethics committees rely on to prove valid consent existed. If the rule were broken, a re-consent after a protocol amendment could be silently overwritten or a withdrawn consent erased, making it impossible to demonstrate which wording a participant actually agreed to and undermining every downstream analysis that assumes valid consent.

I would enforce this at the database level with an **INSTEAD OF UPDATE, DELETE trigger** on `ConsentSigning` that rejects any such operation with an error, while leaving `INSERT` free (see `script/HDS_DB_LAINSBURY_2026.sql`):

```
CREATE OR ALTER TRIGGER dbo.trg_ConsentSigning_AppendOnly
ON dbo.ConsentSigning
INSTEAD OF UPDATE, DELETE
AS
BEGIN
    SET NOCOUNT ON;
    THROW 51000, 'ConsentSigning is append-only: rows cannot be updated or
    ↪ deleted.', 1;
END;
```

This guarantees the rule regardless of which application or user connects, which a permission grant alone would not. For stronger assurance, a SQL Server append-only **ledger table** refuses `UPDATE/DELETE` at the engine level and chains rows into a cryptographically verifiable hash tree, so even an administrator cannot tamper with a consent record undetectably — stronger than the trigger, which a sufficiently privileged account could disable.

Chapter 4

Reference

Appendix A — Data dictionary

A complete column-level reference for every table, derived from the `CREATE TABLE` section of `script/HDS_DB_LAINSBURY_2026.sql`. *Key*: PK = primary key, FK = foreign key, UK = unique (business) key. All surrogate `*_id` keys are `INT IDENTITY(1,1)`.

This dictionary is not static documentation: the metadata section of `script/HDS_DB_LAINSBURY_2026.sql` stores every table description — and the column descriptions where meaning is not self-evident, in particular the meaningful-NULL columns — **in the database catalog itself** as `MS_Description` extended properties. The dictionary can therefore be regenerated at any time (or consumed by data-catalogue tooling) by querying the system views, so the metadata travels with the schema rather than living in a document that can drift out of date:

```
SELECT t.name AS table_name,
       c.name AS column_name,
       TYPE_NAME(c.user_type_id) AS data_type,
       IIF(c.is_nullable = 1, 'NULL', 'NOT NULL') AS nullability,
       CAST(ep.value AS NVARCHAR(400)) AS column_description
FROM sys.tables t
JOIN sys.columns c ON c.object_id = t.object_id
LEFT JOIN sys.extended_properties ep
      ON ep.major_id = c.object_id AND ep.minor_id = c.column_id
      AND ep.class = 1 AND ep.name = 'MS_Description'
ORDER BY t.name, c.column_id;
```

ClinicalTrial

Table 4.1: ClinicalTrial

Column	Type	Null	Key	Notes
<code>trial_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>registration_number</code>	VARCHAR(20)	NOT NULL	UK	ISRCTN number; externally recognised natural key.
<code>trial_name</code>	NVAR- CHAR(150)	NOT NULL		Full trial title.
<code>phase</code>	VARCHAR(4)	NOT NULL		CHECK IN ('I','II','III','IV').
<code>status</code>	VARCHAR(20)	NOT NULL		CHECK IN ('recruiting','active','completed','suspended').

Column	Type	Null	Key	Notes
start_date	DATE	NOT NULL		Trial start date.
end_date	DATE	NULL		NULL = ongoing. CHECK (end_date >= start_date).

HospitalSite

Table 4.2: HospitalSite

Column	Type	Null	Key	Notes
site_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
site_code	VARCHAR(10)	NOT NULL	UK	Short operational code, e.g. WBK-GEN.
site_name	NVAR- CHAR(100)	NOT NULL		Hospital site name.
city	NVAR- CHAR(60)	NOT NULL		City.
country	NVAR- CHAR(60)	NOT NULL		Country.

Participant

Table 4.3: Participant

Column	Type	Null	Key	Notes
participant_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
study_code	VAR- CHAR(15)	NOT NULL	UK	Pseudonym; only external identifier (no names held).
date_of_birth	DATE	NOT NULL		CHECK (date_of_birth < registration_date).
sex_at_birth	VAR- CHAR(10)	NOT NULL		CHECK IN ('Male','Female'). Records biological sex at birth, not gender identity — only two values are captured for this field.
registration_date	DATE	NOT NULL		Date first registered with the research office.

InactivationReason

Table 4.4: InactivationReason

Column	Type	Null	Key	Notes
reason_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
reason_code	VARCHAR(30)	NOT NULL	UK	Standardised report token, e.g. consent_withdrawn.

Column	Type	Null	Key	Notes
reason_description	NVAR-CHAR(255)	NOT NULL		Human-readable description of the reason.

TrialSite (link table)

Table 4.5: TrialSite. The only table in the schema without a surrogate IDENTITY key — the (trial_id, site_id) pairing is itself the row’s identity.

Column	Type	Null	Key	Notes
trial_id	INT	NOT NULL	PK, FK	-> ClinicalTrial . Part of the composite PK (trial_id, site_id).
site_id	INT	NOT NULL	PK, FK	-> HospitalSite . Part of the composite PK (trial_id, site_id).
opened_date	DATE	NOT NULL		Date the site opened for recruitment on this trial.
closed_date	DATE	NULL		NULL = still open. CHECK (closed_date >= opened_date).

ConsentVersion

Table 4.6: ConsentVersion

Column	Type	Null	Key	Notes
consent_version_id	INT	NOT NULL	PK	Surrogate primary key.
trial_id	INT	NOT NULL	FK, UK	-> ClinicalTrial . Part of UNIQUE (trial_id, version_number).
version_number	INT	NOT NULL	UK	CHECK (version_number > 0). Part of the unique key.
effective_from	DATE	NOT NULL		Date this version came into effect.
wording_text	NVAR-CHAR(MAX)	NOT NULL		Full consent-form wording; never overwritten.

Enrollment

Table 4.7: Enrollment. A filtered unique index allows only one active enrolment per participant (WHERE inactivation_date IS NULL).

Column	Type	Null	Key	Notes
enrollment_id	INT	NOT NULL	PK	Surrogate primary key.
participant_id	INT	NOT NULL	FK, UK	-> Participant . Part of UNIQUE (participant_id, trial_id, site_id).
trial_id	INT	NOT NULL	FK, UK	Composite FK with site_id -> TrialSite (confirms the site actually runs the trial). Part of the unique key.

Column	Type	Null	Key	Notes
site_id	INT	NOT NULL	FK, UK	Composite FK with trial_id -> TrialSite. Part of the unique key.
enrolment_date	DATE	NOT NULL		Date enrolment began.
inactivation_date	DATE	NULL		NULL = still active. CHECK (inactivation_date >= enrolment_date).
inactivation_reason_id	INT	NULL	FK	-> InactivationReason. Paired-null CHECK with inactivation_date.

ConsentSigning (append-only)

Table 4.8: ConsentSigning. Protected by an INSTEAD OF UPDATE, DELETE trigger so rows can only ever be inserted.

Column	Type	Null	Key	Notes
consent_signing_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
enrollment_id	INT	NOT NULL	FK, UK	-> Enrollment. Part of UNIQUE (enrollment_id, consent_version_id).
consent_version_id	INT	NOT NULL	FK, UK	-> ConsentVersion. Part of the unique key.
signed_date	DATE	NOT NULL		Date the consent was signed.
witnessed_by	NVAR- CHAR(100)	NOT NULL		Name/role of the witnessing staff member.

Appendix B — File manifest

The submission is a single, self-contained script, run top to bottom:

Table 4.9: `script/HDS_DB_LAINSBURY_2026.sql` — section contents in run order

Section (in run order)	Contents
Create database	(Re)creates the <code>HDS_ClinicalTrial</code> database
Create tables	<code>CREATE TABLE</code> s, constraints, indexes
Insert reference data	The five reference sheets from the brief
Insert additional operational data	<code>TrialSite</code> , <code>Enrollment</code> , <code>ConsentSigning</code> data
Append-only trigger	<code>INSTEAD OF UPDATE</code> , <code>DELETE</code> enforcement on <code>ConsentSigning</code>
Queries	The seven queries
Metadata	Extended-property metadata + catalog-driven data dictionary

Appendix C — Interoperability mapping (HL7 FHIR)

The registry does not exist in isolation: if the trust later needs to exchange trial data with other NHS organisations, sponsors, or national research platforms, the natural target is the HL7 FHIR standard used across the NHS. The schema maps cleanly onto FHIR resources, which validates the entity choices and gives a ready-made route for future data linkage beyond the trust:

Table 4.10: Mapping of registry tables to HL7 FHIR resources

Registry table	FHIR resource	Notes
ClinicalTrial	ResearchStudy	Phase and status map to the resource's phase and status elements.
Participant	Patient (pseudonymised)	study_code becomes a pseudonymous identifier ; no name elements populated.
Enrollment	ResearchSubject	FHIR likewise links one patient to one study with period and status.
HospitalSite	Location / Organization	Site code as the identifier .
ConsentSigning + ConsentVersion	Consent	FHIR Consent also references the specific policy/form version agreed to.
InactivationReason	ResearchSubject progress reason (coded)	The controlled vocabulary would be published as a FHIR CodeSystem .

Coversheet

MSt Healthcare Data Science — Authentication of practice

I confirm that I have fully read and understood the assignment brief for this module. **Y**

Details

Name	Malik J. Lainsbury
Submission Date	03 – 07 – 26
Word Count: whole assignment including codes	~2,900 word-equivalents (1,447 prose + 7 queries x 150 + 8 INSERT statements x 50)
Word Count: main body excluding abstract, references and supplementary materials	1,447 prose words (appendices excluded as supplementary)

Permission to share your assignment

I do give permission to share my assignment with future MSt participants.

University statement of originality

This assignment is the result of my own work and includes nothing which is the outcome of work done in collaboration except as declared in the Preface and specified in the text. It is not substantially the same as any that I have previously submitted for a degree or diploma or other qualification at the University of Cambridge or any other university or similar institution, or that is being concurrently submitted, except as declared in the Preface and specific in the text. I further state that no substantial part of my Portfolio has already been submitted, nor is being concurrently submitted for any such degree, diploma or other qualification at the University of

Cambridge or any other university or similar institution except as declared in the Preface and specified in the text.

I confirm the statement of originality as above **Y**

Questions for reflection

1. Which aspects of this assignment are you most uncertain about and/or would most like to receive feedback on? Whether giving `Enrollment` a composite FK (`trial_id`, `site_id`) straight to `TrialSite` (rather than a single surrogate `trial_site_id` pointer) is considered good practice or over-normalisation; and whether the filtered unique index is an acceptable way to express the “one active trial at a time” rule.

5. Using the wording in the rubric, how would you describe the quality of the different aspects of your work? The ERD, table descriptions and relationships aim at the “high performance” descriptors: all entities identified, the many-to-many resolved with a link table, and each decision justified against the scenario. The script runs end-to-end without errors and every foreign key is resolved by subquery on a business key.

Appendix D — Additional Operational Data

4.1 Overview

Assignment section 6.1 specifies: “In addition to the reference data provided, you must also add **your own data** to all the other records you have designed and there should be **enough data of your own to answer all seven queries meaningfully**.”

The reference data provided (ClinicalTrial, HospitalSite, InactivationReason, Participant, ConsentVersion) contains no TrialSite, Enrollment, or ConsentSigning rows at all — those three tables are entirely self-authored operational data. Beyond the minimum needed to satisfy the six worked-example requirements, two further **trial transitions** were added to enrich Queries 4, 5 and 7. TrialSite uses a composite primary key (trial_id, site_id) rather than a surrogate IDENTITY (see Table Descriptions, main body); as a result Enrollment stores trial_id and site_id directly rather than a single trial_site_id pointer.

4.2 Why a transition, not a second simultaneous enrolment

The schema enforces the brief’s rule that a participant may be actively enrolled in only one trial at a time with a filtered unique index:

```
CREATE UNIQUE INDEX UX_Enrollment_OneActivePerParticipant
ON dbo.Enrollment(participant_id)
WHERE inactivation_date IS NULL;
```

This permits at most one row per participant with inactivation_date IS NULL. Consequently, the two additions below each **close** the participant’s existing active enrolment first (an UPDATE in place, preserving the original row as the brief requires), then **insert** a new active enrolment — exactly the “left a previous trial and later joined a new one” pattern the scenario itself describes.

4.3 Additional Trial Transitions Added

4.3.1 1. WBK-002: CARDIOPROTECT (St Katherine’s) closed, BREATHE (Reading Community) opened

Data added: - UPDATE: WBK-002’s active CARDIOPROTECT enrolment at St Katherine’s is closed — `inactivation_date = 2025-02-15`, `inactivation_reason_id = eligibility_lost` - INSERT: new active enrolment — BREATHE at Reading Community, `enrolment_date = 2025-03-01` - INSERT: consent signing — BREATHE v1, signed 2025-03-01, witnessed by Nurse S. Patel - **Significance:** this is the first and only enrolment ever recorded at Reading Community, activating a previously empty site

Impact on queries: - **Query 5:** Reading Community moves from 0 to 1 total enrolment — all three sites now appear with a non-zero count, demonstrating the `LEFT JOIN` correctly includes a site the moment it gains its first enrolment - **Query 4:** WBK-002 still appears (now under BREATHE/Reading Community instead of CARDIOPROTECT/St Katherine’s) — the active count is unchanged, but site/trial diversity increases - **Query 7:** Reading Community now has a candidate row, so all three sites are represented in the “most recent per site” result

4.3.2 2. WBK-005: BREATHE (Westbrook General) closed, CARDIOPROTECT (St Katherine’s) opened

Data added: - UPDATE: WBK-005’s active BREATHE enrolment at Westbrook General is closed — `inactivation_date = 2025-03-15`, `inactivation_reason_id = clinician_decision` - INSERT: new active enrolment — CARDIOPROTECT at St Katherine’s, `enrolment_date = 2025-04-01` (the most recent enrolment date in the whole database) - INSERT: consent signing — CARDIOPROTECT v2 (the current version), signed 2025-04-01, witnessed by Nurse A. Okafor

Impact on queries: - **Query 7:** gives St Katherine’s a genuine choice between two dated candidates (WBK-002’s original 2025-01-10 enrolment and WBK-005’s 2025-04-01 enrolment), so the `ROW_NUMBER() OVER (PARTITION BY site ...)` logic is doing real selection work rather than returning a trivial single-row partition - **Query 4:** WBK-005 still appears (now under CARDIOPROTECT/St Katherine’s instead of BREATHE/Westbrook General)

4.4 Data Coverage Summary

Metric	Before additions	After additions
Enrollment rows	7	9
Active enrolments	4	4 (unchanged — each transition closes one, opens one)
Hospital sites with \$ \$1 enrolment ever	2	3 (Reading Community now included)
Consent signing events	8	10

Metric	Before additions	After additions
Trials each active participant has moved between	0	2 participants (WBK-002, WBK-005) demonstrate the “left one trial, joined another” pattern

4.5 Query 4, 5 and 7 results after the additions

Query 4 (active enrolments, 4 rows): WBK-002 / BREATHE / Reading Community / 2025-03-01; WBK-001 / CARDIOPROTECT / Westbrook General / 2023-03-15; WBK-003 / CARDIOPROTECT / Westbrook General / 2024-10-01; WBK-005 / CARDIOPROTECT / St Katherine’s / 2025-04-01.

Query 5 (enrolments per site, 3 rows): Westbrook General 4; St Katherine’s 4; Reading Community 1.

Query 7 (most recent per site, 3 rows): Reading Community — WBK-002 / BREATHE / 2025-03-01; St Katherine’s — WBK-005 / CARDIOPROTECT / 2025-04-01; Westbrook General — WBK-003 / CARDIOPROTECT / 2024-10-01.

1. Art. 5 GDPR – Principles relating to processing of personal data. *General Data Protection Regulation (GDPR)*.
2. The Caldicott Principles. *GOVUK*.